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Entropy-Based Modeling of Resilient Manufacturing Systems

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Abstract: Manufacturing resilience is strongly influenced by the ability of production systems to maintain operational continuity under stochastic equipment failures and constrained maintenance capacity. Building on an entropy-based stochastic framework, a modeling approach was developed to quantify operational uncertainty and characterize the resilience of a manufacturing system comprising M identical and independently operating machines supported by a single maintenance crew. Machine failures and repairs were assumed to occur randomly at specified failure and repair rates, respectively, while no more than one failed machine could be repaired at any given time. The resulting machine repair system was formulated as a birth-death process of finite size $M + 1$, in which each state was defined by the number of operational machines. Steady-state probabilities were derived as functions of the repair rate-to-failure rate ratio and were subsequently used to quantify information entropy, the expected numbers of operational and failed machines, maintenance crew utilization, and system operational availability. Information entropy was employed as a quantitative measure of uncertainty in the distribution of system states and, consequently, as an indicator of changes in operational resilience. An information stiffness parameter was further introduced. Then entropy hysteresis was identified, and its hysteresis width was interpreted as a measure of information asymmetry associated with transitions between operating regimes. The entropy hysteresis width increased with the number of available machines, M. By integrating stochastic machine-repair dynamics with information-theoretic measures, the proposed framework provides a quantitative basis for evaluating the relationships among equipment reliability, maintenance capacity, operational uncertainty, and manufacturing resilience. The resulting measures can support maintenance capacity planning, downtime mitigation, operational cost control, and production-delivery management, thereby facilitating more resilient and sustainable manufacturing operations.

Keywords: System information and entropy; Uncertainty; Risk; Birth-death process; Resilient management

1 Introduction

In modern industrial environments, supply chain volatility, fluctuating demand patterns, and unexpected operational disruptions present continuous challenges to traditional production systems. Maintaining competitiveness requires moving beyond rigid operational paradigms toward resilient manufacturing management, i.e., systems engineered not only to withstand shocks but also to rapidly adapt, recover, and reconfigure. By integrating advanced stochastic modeling, real-time decision frameworks, and data-driven insights, contemporary production environments can mitigate vulnerabilities across multi-stage processes and machine-to-machine interfaces. The manufacturing companies frequently rely on queuing models to calculate the capacity required for timely order fulfillment. While queuing analysis remains a cornerstone of logistics and supply chain management, its deployment at a granular level, such as synchronizing specific machine-to-machine interfaces, is rarely observed.

Recent studies on multi-stage manufacturing utilize advanced queuing models to account for complex waiting times, service rate variabilities, and routing between diverse machine tools [1]. To tackle real-time operational hurdles, recent literature suggests employing Markov decision processes for dynamically scheduling urgent orders within high-mix, low-volume production settings [2]. Furthermore, Feng and Wang [3] developed a remanufacturing supply chain framework involving both manufacturers and retailers. Accurately modeling complex manufacturing processes remains a major obstacle; consequently, Abhilash et al. [4] introduced an innovative hybrid framework combining physics-based models with explainable artificial intelligence to mitigate modeling errors. Despite significantly enhancing intelligent manufacturing, artificial intelligence applications continue to encounter adoption barriers primarily due to the scarcity of high-quality training data. Conversely, the rise of large-scale foundation models has revolutionized multi-modal artificial intelligence applications. Although these models exhibit efficient data utilization across various sectors, their implementation in intelligent manufacturing remains in its infancy [5].

This study considers a scenario where a manufacturing system consists of M identical machines that operate independently of one another. During operation, the machines break down randomly, characterized by the failure rate of a single machine. There is a maintenance crew that services the machines, but at any given time, it cannot repair more than one machine, characterized by the repair rate of a single machine. This study models the system (i.e., a cell of machines working in parallel) as a birth-death process of finite size $M+1$ with states n, where $n = 0, 1, 2, \dots, M$. In this study, the following three metrics are used: (i) System utilization (ρ), defined as the ratio of the primary birth and death rates of the birth-death process modeling the system, $\rho = \lambda/\mu$, which, physically, determines the (equilibrium) macrostates of the system. (ii) System information stiffness (ν), which is independent of the choice of ρ-scale, reflecting the resilience of the production flow to utilization fluctuations. (iii) System size (M), representing the number of available machines. The metric M connects the metrics ρ and ν. Furthermore, the current number N of operational machines in the system, $N \in \{0, 1, 2, \dots, M\}$, determines the particular states of the system.

The primary goal of this study is to provide a basic understanding of the uncertainty (i.e., risk) associated with the machine repair problem, which can be helpful in designing and managing manufacturing systems. This approach is proposed to reduce the work-in-progress flow of the system to two measures: system information and its expected value, system entropy. The information i owned by the system (i.e., carried by the system's random variable N) is a key feature of the particular states of the birth-death process, and, in essence, the system's entropy $S = E(i)$ reflects the uncertainty inherent in the system. This study relies on Gibbs' entropy formula [6], utilizing Shannon's adaptation for random variables [7], with the analysis restricted to discrete random variables. Thus, entropy is a quantitative measure of expected information, or average surprise. Further, risk associated with the system is the expected loss, and there must be a loss function mapping states to consequences. If the information possessed by the system in a particular state is taken as the loss associated with observing the particular state, then we can identify risk and uncertainty, associated with the system, using the concept of entropy as their natural measure.

In particular, this information-based analysis could turn out to be a scalable technique to describe some unexpected behavior of operational machines resulting in their breaking down. Thus, an effective response to a production surge would be directly related to an efficient management of the risk associated with the manufacturing system. Entropy-based measures have also been employed to quantify operational uncertainty in inventory and supply-chain systems [8]. Papadopoulos et al. [1] used the basic infinite-capacity $M/M/1$ queueing model to determine capacity in both automated assembly units and specialized fabrication cells. Meanwhile, Curry and Feldman [9] used the finite-capacity $M/M/1/M$ queueing model for dimensioning industrial buffer sizes and machine work cells. While many approaches to the stochastic modeling of manufacturing systems have been evaluated [10, 11], analytical models based on system information and system entropy analysis are usually infrequent. In this context, there exists only the widely exploited information-theoretic approach based on the maximum entropy principle given certain constraints, i.e., by use of the Lagrange multipliers method [12–18].

Given the capital investment constraints that exist in many manufacturing facilities, the type of analysis promoted in this study can be a valuable tool in utilizing resources in the most cost-effective way, simply by identifying the optimal values for the parameters: ρ, ν, and M. Given the total system capacity M, the specific value $\rho_{M,\max}$ of the utilization parameter ρ determines the system maximum uncertainty point (then, $\nu = 0$). Specifically, adjusting the ρ value directly allows us to mitigate the system uncertainty (i.e., risk). Furthermore, the expected number of operational and broken machines, maintenance crew utilization, and system operational availability are determined with respect to the system utilization parameter ρ, contributing to the system resilience. By introducing the information stiffness parameter ν, the phenomenon of entropy hysteresis emerges, the width of which indicates the degree of information asymmetry within the system. As the number M of available machines increases, the entropy hysteresis width increases. The motivation for this research is that the measures of entropy (as a function of utilization) and entropy hysteresis (as a function of information stiffness) should be integrated into the optimization and risk-containment strategies of any machine repair problem.

Knowing when to scale up the repair crew capacity within the entire manufacturing process is essential for the uninterrupted operation of the system and a longer lifespan of the machines. This is done by selecting an optimal operating point of the system utilization (i.e., of the information stiffness of the system). Specifically, operating within the lower entropy regime ensures sustained continuity and optimized repair crew deployment, which translates to the following:

- Extended machine lifecycle: Preventing frequent, uncontrolled catastrophic failures reduces mechanical stress and premature component degradation, directly extending operational life and reducing capital waste.
- Reduced energy consumption: Unplanned machine breakdowns often necessitate high-energy startup cycles, auxiliary idling, and reprocessing of defective batches. Maintaining stable system macrostates minimizes these

energy-intensive inefficiencies.

- Resource and material efficiency: Minimizing unexpected disruptions curbs raw material scrap rates and production downtime, optimizing resource utilization across the manufacturing lifecycle.

The rest of this study is structured as follows. Section 2 presents the system model, describing the system performance measures. In Section 3, a system entropy analysis is given and the system is analyzed with respect to its parameters. Section 4 concludes the study.

2 System Model

A system consists of M identical machines that operate independently of one another. During operation, the machines break down randomly, and $\mu > 0$ characterizes the failure rate of a single machine. There is a maintenance crew that services the machines, but at any given time, it cannot repair more than one machine. In this context, $\lambda > 0$ is the repair rate of a single machine. Let N be a random variable denoting the number of machines currently operating, and ${}^{M}N = M - N$ be a random variable denoting the number of machines broken. In other words, ${}^{M}N = n$ if and only if $N = (M - n)$, $n = 0, 1, \ldots, M$. This study supposes that the repair times follow an exponential distribution, while the failure times are independent and identically distributed; under the insensitivity property of loss systems, they may follow an identical general distribution, as discussed in [19]. In this research, this machine repair problem is modeled as a birth-death process of size $M + 1$, i.e., state space $S = \{0, 1, 2, \ldots, M\}$. A state $n \in S$ means there are n machines currently operating and $M - n$ machines broken (that is, waiting for or under repair).

The state changes due to two types of events: machine failures and repairs. Through the failure rates, a state decreases: $n \to n - 1$. When n machines are operating, each breaks down independently at rate μ. The total system failure rate is:

$$\mu_n = n \cdot \mu, \quad n = 1, 2, \ldots, M \tag{1}$$

Through the repair rates, a state increases: $n \to n + 1$. When $n < M$, there are $M - n$ broken machines. Since there is only one repair crew, only one machine can be repaired at a time, regardless of how many are broken. The repair rate is:

$$\lambda_n = \lambda, \quad n = 0, 1, \ldots, M - 1 \tag{2}$$

Let $p_n = \mathrm{Prob}\{N = n\}$ (with $n = 0, 1, \ldots, M$) be the stationary probability that exactly n machines are working. Then, the variable N defines the particular states of the system, and the probability p_n is determined by the probability continuity law (i.e., global balance equations) and probability conservation law (i.e., normalization factor) [20–22].

If λ and μ are considered as the primary birth/death rates of the birth-death process, i.e., primary repair and failure rates (meaning that λ_n/λ and μ_{n+1}/μ, with $n = 0, 1, \ldots, M - 1$, are functions of n only and are independent of λ and μ), then,

$$\rho = \lambda/\mu = \lambda \cdot T_s \tag{3}$$

where, $T_s = 1/\mu$ is a machine failure time, i.e., the time period a machine takes to break down, and ρ is the system utilization parameter (i.e., system working intensity). Now, the stationary probability distribution p_n, with $n = 0, 1, .., M$, is given as:

$$\begin{gathered}
\frac{p_n(\rho)}{p_0(\rho)} = \prod_{j=0}^{n-1} \frac{\lambda_j}{\mu_{j+1}} = \prod_{j=0}^{n-1} \frac{\lambda}{(j+1) \cdot \mu} = \frac{\rho^n}{n!}, \quad n = 1, 2, \ldots, M, \\
p_0(\rho) = \frac{1}{f_M(\rho)}, \quad f_M(\rho) = \sum_{n=0}^{M} \frac{\rho^n}{n!}, \quad \rho > 0,
\end{gathered} \tag{4}$$

Thus, a family of birth-death processes of size $M + 1$ can be obtained, indexed by the parameter ρ, which characterizes the machine failure/repair mode and, consequently, the corresponding family of systems. Physically, a given system can be regarded as a family of systems, determined by the polynomial $f_M(\rho)$, where every value ρ' of the parameter ρ defines one (equilibrium) macrostate of the system, whereby the quantity N follows the probability distribution $p_n(\rho'), n = 0, 1, \ldots, M$.

Therefore, for the probability p_N of N operational machines (i.e., for any state n), the following can be derived:

$$p_n(\rho) = \frac{\rho^n/n!}{\sum_{j=0}^{M} \rho^j/j!}, \quad n = 0, 1, \ldots, M, \quad \rho > 0 \tag{5}$$

The quantity N follows a truncated Poisson distribution of size $M+1$ and parameter ρ. Staying in any of its particular states n, a given system possesses a quantity of information $i_n(\rho)$ (i.e., the variable N carries information $i_N(\rho)$), with possible values as follows:

$$i_n(\rho) \stackrel{\text{def}}{=} -\ln p_n(\rho), \quad n=0,1,\ldots,M, \quad \rho>0 \tag{6}$$

Eq. (5) and Eq. (6) lead to the following:

$$i_n(\rho) = i_0(\rho) + (-\ln\rho)\cdot n + \ln(n!), \quad n=0,1,\ldots,M, \quad \rho>0 \tag{7}$$

Furthermore, for the probability $\left({}^{M}p\right)_N$ of ${}^{M}N=(M-N)$ broken machines (i.e., for any state $M-n$), the following can be derived:

$$\left({}^{M}p\right)_n(\rho) = p_{M-n}(\rho), \quad n=0,1,\ldots,M, \quad \rho>0 \tag{8}$$

The variable ${}^{M}N=(M-N)$ carries information $\left({}^{M}i\right)_N$, with possible values as follows:

$$\left({}^{M}i\right)_n(\rho) = i_{M-n}(\rho), \quad n=0,1,\ldots,M, \quad \rho>0 \tag{9}$$

By substituting Eq. (7) into Eq. (9), the following can be obtained:

$$\left({}^{M}i\right)_n(\rho) = \left({}^{M}i\right)_0(\rho) + (\ln\rho)\cdot n + \ln\frac{(M-n)!}{M!}, \quad n=0,1,\ldots,M, \quad \rho>0 \tag{10}$$

Eq. (8) and Eq. (9) directly lead to the following:

$$S(\rho) = E\left(i_N(\rho)\right) \stackrel{\text{def}}{=} \sum_{n=0}^{M} i_n(\rho)\cdot p_n(\rho) = E\left(\left({}^{M}i\right)_N(\rho)\right) \stackrel{\text{def}}{=} \sum_{n=0}^{M} \left({}^{M}i\right)_n(\rho)\cdot\left({}^{M}p\right)_n(\rho), \quad \rho>0 \tag{11}$$

Thus, the direct system (concerning operational machines N) and the inverse system (concerning broken machines ${}^{M}N=(M-N)$) have the same entropy. Clearly, both families of systems, indexed by the parameter ρ, concern the same manufacturing system of M machines, which is a subject of investigation. Therefore, entropy is a quantitative measure of expected information, or average surprise, and it quantifies the spread of a probability distribution across the particular states. It measures the uncertainty associated with the system without caring about the payload or severity of the outcomes. High entropy means that the outcome is difficult to predict; zero entropy means that the outcome is deterministic. On the other hand, risk is the expected loss or potential for adverse consequences associated with a particular state of the system. It requires a decision-making frame, i.e., there must be a loss function mapping states to consequences. From a decision-theoretic perspective, if the information possessed by the system in a particular state is taken as the loss associated with observing the particular state, then risk and uncertainty, associated with the system, can be equalized using the concept of entropy as their entirely innate measure.

2.1 System Metrics

The expected number of operational machines (i.e., average workload) in a given system is given by:

$$\bar{N}(\rho) \stackrel{\text{def}}{=} \sum_{n=0}^{M} n\cdot p_n(\rho) = \rho\cdot\frac{f'_M(\rho)}{f_M(\rho)} = \rho\cdot\left[i_0(\rho)\right]'_\rho, \quad \rho>0 \tag{12}$$

and,

$$\lim_{\rho\to 0^+}\bar{N}(\rho)=0, \quad \lim_{\rho\to+\infty}\bar{N}(\rho)=M,$$

where, $\bar{N}(\rho)$ is a strictly increasing function of ρ. The following can be obtained from Eq. (4):

$$f'_M(\rho) = f_{M-1}(\rho), \quad \rho>0, \quad (f_0(\rho)=1) \tag{13}$$

Eq. (12) directly leads to the following:

$$\bar{N}(\rho) = \rho\cdot\frac{f_{M-1}(\rho)}{f_M(\rho)} = \rho\cdot\left(1-p_M(\rho)\right), \quad \rho>0 \tag{14}$$

The expected number of broken machines (i.e., average idle-working) in a given system is given by:

$$\overline{{}^{M}N}(\rho) = \overline{(M-N)}(\rho) \stackrel{\text{def}}{=} \sum_{n=0}^{M}(M-n)\cdot p_n(\rho) = M - \bar{N}(\rho)$$
$$\overline{{}^{M}N}(\rho) = M - \rho\cdot\frac{f'_M(\rho)}{f_M(\rho)} = M - \rho\cdot[i_0(\rho)]'_\rho = -\rho\cdot[i_M(\rho)]'_\rho, \quad \rho>0 \tag{15}$$

and,

$$\lim_{\rho\to 0^+}\overline{{}^{M}N}(\rho) = M, \qquad \lim_{\rho\to+\infty}\overline{{}^{M}N}(\rho) = 0$$

where, $\overline{{}^{M}N}(\rho)$ is a strictly decreasing function of ρ. The following can be obtained from Eq. (14):

$$\overline{{}^{M}N}(\rho) = M - \bar{N}(\rho) = M - \rho\cdot(1-p_M(\rho)), \rho>0. \tag{16}$$

Furthermore, there exists a unique value $\rho_{M/2}$ of the parameter ρ, called the half-loading point of the system, such that

$$\bar{N}(\rho)\Big|_{\rho=\rho_{M/2}} = \overline{{}^{M}N}(\rho)\Big|_{\rho=\rho_{M/2}} = M/2. \tag{17}$$

It follows from Eq. (14) that the half-loading point $\rho_{M/2}$ is the unique positive solution of the equation

$$\rho\cdot\frac{f_{M-1}(\rho)}{f_M(\rho)} = \frac{M}{2}$$

Hence, $\rho_{M/2} > M/2, M\geq 1$, and $\frac{\rho_{M/2}}{(M/2)}\to 1$, when $M\to+\infty$.
Thus, the following approximation can be used:

$$\rho_{M/2}\approx\frac{M}{2}, \qquad \text{for} \qquad M\geq 10 \tag{18}$$

The repair crew utilization (i.e., the probability that the repair crew is busy, that is, the crew is busy as long as at least one machine is broken, i.e., $N<M$, or ${}^{M}N>0$) is

$$U(\rho) = 1 - p_M(\rho) = \frac{f_{M-1}(\rho)}{f_M(\rho)} = \frac{\bar{N}(\rho)}{\rho} \tag{19}$$

and,

$$\lim_{\rho\to 0^+}U(\rho) = 1, \qquad \lim_{\rho\to+\infty}U(\rho) = 0$$

where, $U(\rho)$ is a strictly decreasing function of ρ.

The operational availability (i.e., the probability that the system is operating, that is, the system is operating as long as at least one machine is operational, i.e., $N>0$, or ${}^{M}N<M$) is

$$A(\rho) = 1 - p_0(\rho) = \frac{f_M(\rho)-1}{f_M(\rho)} \tag{20}$$

and,

$$\lim_{\rho\to 0^+}A(\rho) = 0, \qquad \lim_{\rho\to+\infty}A(\rho) = 1,$$

where, $A(\rho)$ is a strictly increasing function of ρ.

Hence, for the relationship between crew utilization U and operational availability A of the system, the following can be obtained:

$$A(\rho) = 1 - [1-U(\rho)]\cdot\frac{M!}{\rho^M},$$
$$U(\rho) = 1 - [1-A(\rho)]\cdot\frac{\rho^M}{M!}$$

It follows from Eq. (4) that the point where the crew utilization U and operational availability A of the system are equal, the so-called crossover point for the pair (U, A), is

$$\rho_{(p_0=p_M)}\equiv\rho_{(U=A)} = \sqrt[M]{M!} \tag{21}$$

Using Stirling's formula,

$$M! \approx \sqrt{2 \cdot \pi \cdot M} \cdot \left(\frac{M}{e}\right)^{M}, \text{ for } M \geq 5$$

the following approximation can be obtained:

$$\rho_{(U=A)} \approx \frac{M}{e} \cdot \sqrt[2\cdot M]{2 \cdot \pi \cdot M}\Big|_{M \geq 5} \approx \frac{M}{e}\Big|_{M \geq 200} \tag{22}$$

Thus, Eq. (22) and Eq. (18) yield the following useful approximation:

$$\frac{\rho_{(U=A)}}{\rho_{M/2}} \approx \frac{2}{e} \cdot \sqrt[2\cdot M]{2 \cdot \pi \cdot M}\Big|_{M \geq 10} \approx \frac{2}{e}\Big|_{M \geq 200} \approx 0.736$$

3 System Entropy Analysis

For the entropy of the system, Eq. (11) leads to the following:

$$S(\rho) = \ln f_M(\rho) - (\ln \rho) \cdot \bar{N}(\rho) + \frac{1}{f_M(\rho)} \cdot \sum_{n=1}^{M} \left(\frac{\ln n!}{n!}\right) \cdot \rho^n, \quad \rho > 0 \tag{23}$$

Eq. (23) also presents the relationship between the system entropy $S(\rho)$ and its expected number $\bar{N}(\rho)$ of operational machines.

The function $S(\rho) \to 0$, as $\rho \to 0^+$or $\rho \to +\infty$, and attains its maximum value in the maximum uncertainty point $\rho_{M,\max}$ of parameter ρ, which is the unique positive solution of the equation (obtained as $S'(\rho) = 0$) :

$$\sum_{n=0}^{M} \left[n - \rho \cdot \frac{f_{M-1}(\rho)}{f_M(\rho)}\right] \cdot \left[\frac{\rho^n}{n!} \cdot \ln\left(\frac{\rho^n}{n!}\right)\right] = 0 \tag{24}$$

The relationship between the entropy S and utilization ρ of the system is illustrated in Figure 1 (for $M = 5, 10$). The sequence $\{\rho_{M,\max}; M \geq 1\}$ is strictly increasing and unlimited. Using Eq. (24), it can be found that, if $M = 1, 5, 10, 15, 20$, then $\rho_{M,\max} = 1, 2.9, 5.8, 9.0, 12.5$, respectively. In Figure 1, when $\rho \in (0, \rho_{M,\max})$, the system is in part (a) of the entropy $S(\rho)$, where machine failures dominate, and when $\rho \in (\rho_{M,\max}, +\infty)$, the system is in part (b) of the entropy $S(\rho)$, where machine repairs dominate.

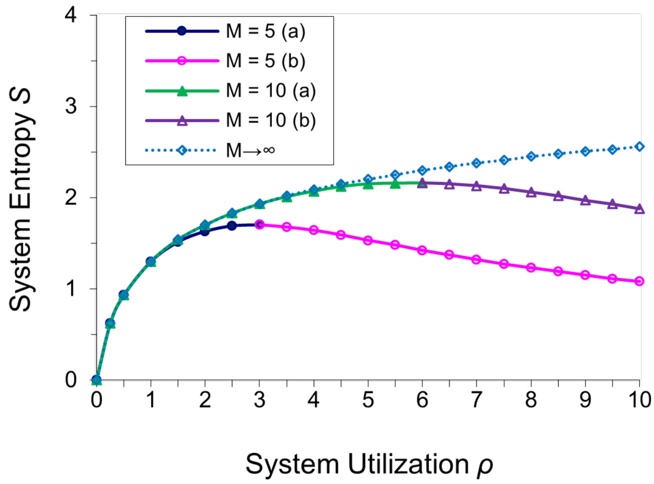

Figure 1. System entropy S vs. system utilization ρ for different system sizes M

Note: $\rho_{M,\max} = 2.9$ for $M = 5$, $\rho_{M,\max} = 5.8$ for $M = 10$, and $\rho_{M,\max} \to +\infty$ as $M \to \infty$.

Furthermore, the following can be obtained for the system:

$$\rho_{(U=A)} < \rho_{M/2} < \rho_{M,\max}$$

The dependence of the points $\rho_{(U=A)}$, $\rho_{M/2}$, and $\rho_{M,\max}$ on the maximum number of machines M (i.e., system size) is shown in Figure 2.

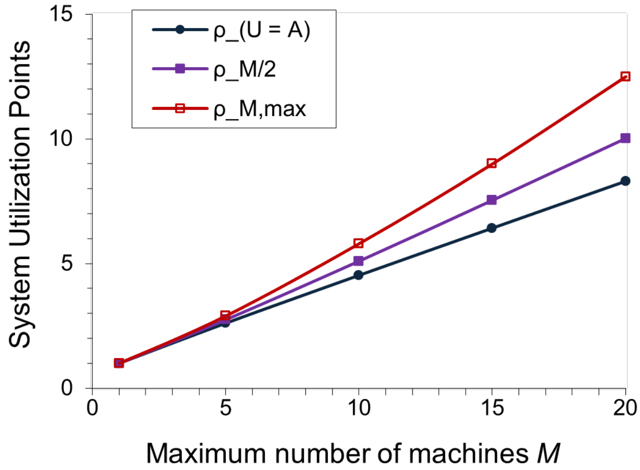

Figure 2. Characteristic utilization points $\rho_{(U=A)}, \rho_{M/2}$, and $\rho_{M,\max}$ vs. system size M

If $\rho \in (0, \rho_{M,\max})$, then Eq. (7) can be rewritten in the following form:

$$i_n(\rho) = i_0(\rho) + \left(-\ln \frac{\rho}{\rho_{M,\max}}\right) \cdot n + \ln \frac{n!}{\rho^n_{M,\max}}, \quad n = 0, 1, \ldots, M \tag{25}$$

Similarly, if $\rho \in (\rho_{M,\max}, +\infty)$, then Eq. (10) can be rearranged in the following form:

$$\left({}^{M}i\right)_n(\rho) = \left({}^{M}i\right)_0(\rho) + \left(\ln \frac{\rho}{\rho_{M,\max}}\right) \cdot n + \ln \frac{(M-n)! \cdot \rho^M_{M,\max}}{M! \cdot \rho^{M-n}_{M,\max}}, \quad n = 0, 1, \ldots, M \tag{26}$$

Therefore, for suitable expression of the system information, an information parameter v of a system can be introduced, which is the second key metric in the proposed methodology. A value v $(v > 0)$ of the information parameter of the given system is related to every pair of values ρ' and ρ'' $(\rho' < \rho'')$ of the system utilization parameter ρ that are inversely symmetrical with respect to the point $\rho_{M,max}$ (i.e., $\rho' \cdot \rho'' = \rho^2_{M,max}$), as follows:

$$v = -\ln \frac{\rho'}{\rho_{M,\max}} = \ln \frac{\rho''}{\rho_{M,\max}} > 0 \tag{27}$$

As ν decreases, the entropy S increases and, in the limiting case, when $\nu = 0$, we have $S = \max S$. Eq. (25) shows that the ground part i_0, elastic (Hooke's) part i_{el} (being proportional to N), and inelastic part or synchronization part i_{syn} of the information i_N of the direct system can differ. Similarly, according to Eq. (26), the ground part $\left({}^{M}i\right)_0$, elastic (Hooke's) part $\left({}^{M}i\right)_{el}$ (being proportional to ${}^{M}N = M - N$), and inelastic part or synchronization part $\left({}^{M}i\right)_{\text{syn}}$ of the information $\left({}^{M}i\right)_N$ of the inverse system can differ. The former two parts form the regular part of the system information, i.e., the part which depends on the parameters ρ and ν. In contrast, the synchronization part of the system information is the part which is not dependent on these parameters. Furthermore, setting ρ' and ρ'' in Eq. (27) yields the following:

$$\begin{aligned} i_{el}(\rho') &= v \cdot N, \\ \left({}^{M}i\right)_{el}(\rho'') &= v \cdot \left({}^{M}N\right) \end{aligned} \tag{28}$$

Parameter v gets a deeper physical interpretation as the information stiffness of a system. To some extent, the inelastic part can be considered as information noise, i.e., some movements in the elastic part.

If $\rho = c \cdot r$, then $\rho_{M,\,\max} = c \cdot r_{M,\,\max}$ (the basic Eq. (24) for $\rho_{M,\,\max}$) and, consequently, the information stiffness v of the system does not depend on ρ-scale $(v = \mp \ln(\rho/\rho_{M,\max}))$. Thus, all terms in Eq. (25) and Eq. (26) do not depend on ρ-scale. But it is not the case with representations in Eq. (7) and Eq. (10).

For a given system of machines, which can function in both regimes (i.e., parts (a) and (b)), the study introduces the following:

$$\begin{gathered} S^{(1)}_M(v) = S(\rho'), \quad S^{(2)}_M(v) = S(\rho''); \\ \rho' < \rho'', \quad \rho' \cdot \rho'' = \rho^2_{M,\max}; \quad v = -\ln \frac{\rho'}{\rho_{M,\max}} = \ln \frac{\rho''}{\rho_{M,\max}} > 0 \end{gathered} \tag{29}$$

Thus, passing from utilization parameter ρ to information stiffness parameter v (i.e., its dynamic version, free of choice of ρ-scale) as a variable, the curve

$$S_M(\rho) \equiv S(\rho), \quad \rho \in (0, +\infty)$$

is geometrically replaced by the two curves,

$$(a) S_M^{(1)}(v)\Big|_{\rho=\rho'}, \quad (b) S_M^{(2)}(v)\Big|_{\rho=\rho''}, \quad v \in (0, +\infty). \tag{30}$$

Clearly,

$$S_M^{(1)}(0) = S_M^{(2)}(0) = \max_{\rho \in (0,+\infty)} S_M(\rho) = S_M(\rho_{M,\max})$$

$$\lim_{v \to +\infty} S_M^{(1)}(v) = \lim_{v \to +\infty} S_M^{(2)}(v) = 0$$

In accordance with Eq. (23), the first curve in Eq. (30), denoted as (a), represents a more risky operating regime, whereas the second curve, denoted as (b), represents a less risky operating regime of the system.

The representation of entropy through information parameter v ($v > 0$), using the variable change in Eq. (29), is presented in Figure 3 for $M = 5$ and in Figure 4 for $M = 10$. Since this system is information-asymmetrical (meaning that the polynomial $f_M(\rho)$ in Eq. (4) is not a symmetrical polynomial), the curves (a) and (b) do not cover each other. In addition, the curve b is lower than the curve a. Then, for any given v ($v > 0$), the system in the point $\rho'' \in (\rho_{M,\max}; +\infty)$ functions with entropy loss or smaller risk with respect to the point $\rho' \in (0; \rho_{M,\max})$, where $\rho' \cdot \rho'' = \rho_{M,\max}^2$. Thus, the interval $(\rho_{M,\max}; +\infty)$ is entropy loss or a smaller risk interval with respect to the interval $(0; \rho_{M,\max})$.

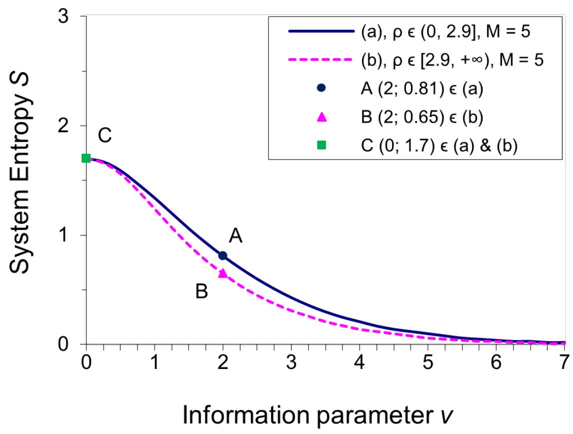

Figure 3. System entropy S versus information stiffness ν for $M = 5$; $A \to B$ via C indicates negative entropy hysteresis

Note: $\rho_{M,\max} = 2.9$.

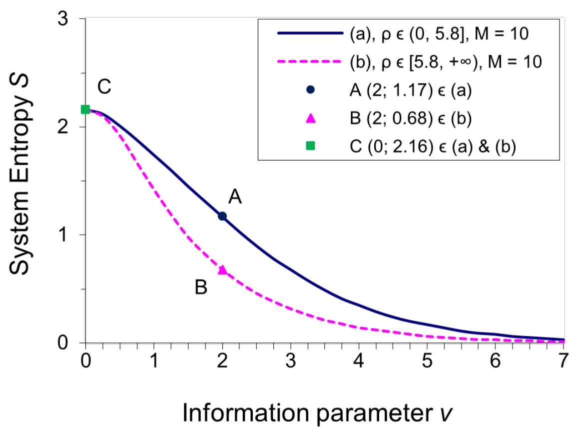

Figure 4. System entropy S versus information stiffness ν for $M = 10$; $A \to B$ via C indicates negative entropy hysteresis

Note: $\rho_{M,\max} = 5.8$.

Let us consider an increase in parameter ρ from ρ' to ρ'' ($\rho' < \rho''$) as a sequence of equilibrium macrostates of the system, evolving from the macrostate ρ' to the macrostate ρ'' via the value $\rho_{M,\max}$. As it can be seen from Figure 3 and Figure 4, when the parameter ρ increases (i.e., $A \to B$ via C), for the system there exists a phenomenon of negative entropy (i.e., risk) hysteresis. According to the two figures, as the system size M increases, the width of the resulting entropy hysteresis increases, as does the degree of information asymmetry of the system. Taking this phenomenon into consideration, especially at larger system sizes (i.e., at a larger number of available machines), can be especially useful when looking at the difference in entropy values by concerning two inversely symmetrical points of the system utilization parameter ρ with respect to the maximum uncertainty point $\rho_{M,\max}$.

Therefore, entropy hysteresis can be defined as a path-dependent lag in how system entropy responds to changes in the system utilization parameter ρ. This can be observed by introducing the information parameter υ. The entropy hysteresis is a very significant phenomenon related to the asymmetry of a system. This phenomenon is absent in an information-symmetrical system. Furthermore, as system size M increases, the width of the entropy hysteresis increases for a given value of the system information stiffness. However, the phenomenon disappears completely when $M \to \infty$, since the information stiffness can no longer be introduced in this case. Therefore, the information asymmetry degree of a system can be expressed by the width of its entropy hysteresis.

We designate this phenomenon "entropy hysteresis" because the system entropy exhibits path dependency with respect to the information stiffness υ. Rather than being a single-valued function of the utilization parameter ρ, the change in entropy depends on whether the system is undergoing a period of entropy ramp-up or ramp-down. During rapid decreases in information stiffness, entropy grows along a transient trajectory, and during subsequent increases, entropy decays along a distinct path. This discrepancy between the forward and backward paths constitutes the path-dependent hysteresis.

3.1 Discussion

The system, when operating in the second regime (i.e., part (b)), operates with lower uncertainty, and as the total number M of available machines is larger, this uncertainty is further reduced compared to operating in the first regime (i.e., part (a)). Of course, if it operates near the maximum uncertainty point $\rho_{M,\text{ max}}$, then it retains the most information on average, and for the system it is equally likely that failures or repairs will dominate, maximizing the cognitive tracking load of the system manager. Thus, the focus of the system analysis is shifted from the point $\rho = \lambda/\mu = 1$, which characterizes the ratio of the repair rate to the failure rate, to the points $\rho_{(U=A)}$ and $\rho_{M/2}$, with particular emphasis on the point $\rho_{M,\max}$, which separates the intervals of higher and lower uncertainty, i.e., risk. As M increases, the point $\rho_{M,\max}$ increasingly escapes from the points $\rho_{(U=A)}$ and $\rho_{M/2}$ (Figure 2). Furthermore, the condition $\rho > \rho_{M,\max}$ is also needed so that the entire system functions more relaxed and sustainable and the machines perform their production resiliently.

Thus, the condition for ρ to exceed the point $\rho_{M,\max}$ is sufficient for the system of M parallel machines to function in order to ensure sustained operational continuity, because it simultaneously ensures that ρ is greater than the other two points: $\rho_{(U=A)}$ (where the system's operational availability becomes greater than the repair crew utilization) and $\rho_{M/2}$ (where the expected number of operational machines becomes greater than M/2). Otherwise, the system manager should increase the repair crew capacity, by increasing the repair rate of the existing crew, which can lead to an increase in the parameter ρ. Thus, in terms of self-renewing production, it is only necessary to check whether the experimentally calculated ρ value (which can be obtained at regular time intervals, e.g., hourly or daily) falls within the interval with lower entropy. If it does not, measures should be taken to accelerate machine repair. Concerning resilient manufacturing management, this study emphasizes how a manufacturing system, consisting of M available independent machines working in parallel, needs to absorb disruptions (i.e., machine failures) and autonomously bounce back to a stable operational macrostate via adaptive repair policies.

The model assumes independent machines working in parallel under controlled conditions, which often ignores messy factory-floor complexities (e.g., cascading failures, shared bottlenecks, variable operator skill levels, and non-exponential repair times). Factory operations are inherently non-stationary, characterized by fluctuating product mixes, variable raw material qualities, and differences in operator skill levels that influence both failure propensities and restoration times. The use of information-theoretic rigidity and entropy constraints offers a significant advantage in this study: because these metrics monitor the macroscopic distribution state rather than micro-level fluctuations, they remain robust indicators of system resilience even when underlying operational parameters drift over time.

To enhance the practical applicability of this framework to modern smart manufacturing, several extensions could be considered:

(i) Shared maintenance queuing networks: Integrating explicit finite-capacity repair crew sub-models directly into the stochastic state-transition diagram to evaluate the impact of crew allocation strategies on overall system resilience.

(ii) Empirical validation: Validating the theoretical entropy bounds and ρ-thresholds using high-frequency telemetry and event-log data collected from industrial shop floors.

(iii) Real-time control integration: Developing closed-loop adaptive control algorithms that utilize real-time entropy measurements to dynamically adjust maintenance resource deployment before critical operational thresholds are breached.

Remark 1. The maximum number of available machines M (i.e., system size) and the parameters ρ and ν, which are connected by the size M, are the three key metrics in this methodology. Therefore, supposing in advance an analysis with $M \to \infty$ (i.e., a system with infinite size), most of the quantities present in systems with finite size are lost. Thus, the following conclusion can be drawn:

If the system is modeled from the outset as having an unlimited number of available machines, i.e., if the birth-death process used to model the system is assumed to have an infinite size ($M \to \infty$), then the following important points are no longer defined: $\rho_{M,\max}$, the maximum uncertainty point; $\rho_{M/2}$, the half-loading point; and $\rho_{(U=A)}$, the crossover point for the pair (U, A). Consequently, the phenomenon of entropy hysteresis can no longer be introduced.

Remark 2. Related entropy-based estimation of the utilization parameter, including its relationship with maximum likelihood estimation, has been investigated previously [23]. Entropy-based estimation provides an alternative approach for parameter identification by considering the information contained in the probability distribution of system states. The present study focuses on applying this framework to a finite-source machine-repair system and analyzing its implications for operational uncertainty, maintenance capacity, and manufacturing resilience.

Remark 3. Several studies have integrated uncertainty into the performance optimization of various systems. For instance, the approach proposed by Lazov [24] has been applied for revenue analysis of parking lots, providing a desired optimal trade-off among the measures full parking lot revenue, parking lot mean revenue, mean normalized square deviation of parking lot revenue from its linear part, and uncertainty (i.e., entropy) of the parking lot with respect to the parking lot utilization parameter. Moreover, besides the uncertainty (i.e., risk) captured by an external observer, the risk captured by an arriving customer and a departing customer in a stochastic service system has also been investigated and contrasted [25]. It can be noted that the current study focuses exclusively on risk from an external observer's perspective.

4 Conclusions

Manufacturing systems are mainly complex due to the inherent variability in both the production demands and the service times required to fulfill those demands. In resilient manufacturing management, decisions regarding the allocation of technical staff and operational machinery are often more critical than in other areas because of high capital expenditure and the effects of bottleneck delays. Therefore, it is imperative that these management decisions rely upon inherent system features, such as system information i, and system entropy $S = E(i)$, representing the uncertainty associated with the system. This analysis demonstrates that effective resilient management, based on the uncertainty (i.e., risk) associated with the synchronization between the machine failure rates and crew repair rates, is crucial to sustainable production. Consequently, the entropy, expected number of operational and broken machines, maintenance crew utilization, and system operational availability are evaluated with respect to the system utilization parameter ρ, contributing to the system resilience. By introducing the information stiffness parameter v, the phenomenon of entropy hysteresis emerges, the width of which indicates the degree of information asymmetry within the system. Since the entropy has a maximum value $S\left(\rho_{M,\max}\right)$, the strategic goal is to operate the system within the lower risk interval, i.e., $\rho \in (\rho_{M,\max}, +\infty)$, compared to the greater risk interval, i.e., $\rho \in (0, \rho_{M,\max})$.

Regarding management objectives, the number of available machines M connects the metrics ρ and v. Thus, the crew repair rate λ and the machine failure rate μ can be experimentally measured for various working scenarios, and then ρ and v can be determined. In this study, adjusting the crew repair rate (by increasing/decreasing the crew capacity) directly allows managers to mitigate system entropy (i.e., uncertainty) and observe its direct correlation with the expected number of operational machines in the system. Effective management requires balancing between machine failures and maintenance crew repairs. To meet these challenges and improve delivery performance while controlling operational costs, production managers can utilize this analytical approach to contain the impact of equipment downtime. Moving forward, this approach can be applied to critical resource management cases as a direction for future research.

Data Availability

The data supporting the research results are included within the article.

Conflicts of Interest

The author declares no conflicts of interest.

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